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Certificate No. 21/0822/2

TO: AstraZeneca

ADDRESS: AKB Building, Granta Park, Cambridge, CB21 6GH

THE SAMPLE OF: 1 x MEDI0382 Cotadutide

RECEIVED FROM YOU ON: 7 September 2021

MARKED: Batch No. SP19-097 In use stability T24m 30C

REQUIREMENT: Antimicrobial Effectiveness Test

METHOD REFERENCE: HMRM/2K/M71/01

DATA REFERENCE: AZ Micro Book 1, RJ

TEST	SPECIFICATION	RESULT
Antimicrobial Effectiveness Test	Complies to the Current Ph.Eur. USP and JP	Complies to the Ph.Eur. 10th Edition (Criteria B), USP-NF 2021 and JP17

Remarks: The analysis contained in this certificate so far as can be reasonably established has been carried out in accordance with cGMP and accurately reflects the raw data generated.

Checked by:

[redacted]

Microbiology Department

Approved by:

[redacted]

Quality Assurance Department

REGISTERED IN ENGLAND NO: 5391062

TABLE 1 – ANTIMICROBIAL EFFECTIVENESS TEST

Sample: 1 x MEDI0382 Cotadutide
Batch No: SP19-097 In use stability T24m 30C
Method Reference: HMRM/2K/M71/01

SUMMARY OF ORGANISM RECOVERIES

	Candida albicans NCPF 3179 (ATCC 10231)		Aspergillus brasiliensis IMI 149007 (ATCC 16404)		Pseudomonas aeruginosa NCIMB 8626 (ATCC 9027)		Staphylococcus aureus NCTC 10788 (ATCC 6538)		Escherichia coli NCIMB 8545 (ATCC 8739)	
	CFU/ml	Log Reduction	CFU/ml	Log Reduction	CFU/ml	Log Reduction	CFU/ml	Log Reduction	CFU/ml	Log Reduction
Aqueous Controls	7.10×10^5 (log 5.9)	N/A	2.45×10^5 (log 5.4)	N/A	1.85×10^5 (log 5.3)	N/A	2.80×10^5 (log 5.4)	N/A	2.80×10^5 (log 5.4)	N/A
0 hours	3.10×10^5	0.4	1.65×10^5	0.2	1.36×10^5	0.2	1.38×10^5	0.3	1.66×10^5	0.2
6hours	1.36×10^5	0.8	3.05×10^4	0.9	<10	>4.3	1.76×10^4	1.2	<10	>4.4
24hours	7.90×10^3	2.0	9.50×10^1	3.4	<10	>4.3	2.20×10^2	3.1	<10	>4.4
7days	<10	>4.9	<10	>4.4	<10	>4.3	<10	>4.4	<10	>4.4
14days	<10	>4.9	<10	>4.4	<10	>4.3	<10	>4.4	<10	>4.4
28days	<10	>4.9	<10	>4.4	<10	>4.3	<10	>4.4	<10	>4.4

N/A = Not Applicable

Checked by: [redacted] Microbiology Department

Approved by: [redacted] Quality Assurance Department

ACCEPTANCE CRITERIA

PH.EUR. REQUIREMENT

Parenteral preparations, eye preparations, intrauterine preparations and intramammary preparations

		Log Reduction				
		6 h	**24 h**	**7 d**	**14 d**	**28 d**
Bacteria	**A**	2	3	-	-	No Recovery
	B	-	1	3	-	No Increase
Fungi	**A**	-	-	2	-	No Increase
	B	-	-	-	1	No Increase

The A criteria express the recommended efficacy to be achieved. In justified cases where the A criteria cannot be attained such as a potential of an increased risk of adverse reaction, the B criteria must be satisfied.

No Increase is no increase in number of viable micro-organisms compared to the previous reading.

USP-NF REQUIREMENT

Products are divided into 4 categories, Cotadutide Drug Product is a category 1 (Refer to table below).

JP REQUIREMENT

Products are divided into 4 categories, Cotadutide Drug Product is a category 1A (Refer to table below).

For Category 1 and 1A Products	
Bacteria	At 7 days: Not less than 1.0 log reduction from the initial count At 14 days: Not less than 3.0 log reduction from the initial At 28 days: No increase from the 14 days
Yeast and Moulds	At 7, 14 and 28 days no increase from the initial count

No increase from the initial count means not more than 0.5 $\log_{10}$ increase from the initial calculated count.